Research Proposal: The Title

Student ID: 2250500xx

Student Name: guwudx

Mentor: Prof. Yongfei Wang

Course Instructor:: Prof. Yongfei Wang

Department: School of Medicine, MSC in Bioinformatics

Institution: The Chinese University of Hong Kong, Shenzhen

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A. Background and Significance

This section will provide a comprehensive overview of the current state of research in the field, highlighting the gaps and challenges that our proposed study aims to address. We will discuss the significance of our research question and how it contributes to advancing knowledge in the area. Additionally, we will review relevant literature to establish the context for our specific aims and research design. We will also emphasize the potential impact of our findings on both the scientific community and clinical practice, unders ...

B. Specific Aims

The long-term objective of this research is to establish.... To test the central hypothesis that ... , we propose the following specific aims:

- **Aim 1: To ...** We will test/construct/verify a ...
- **Aim 2: To ...** We will implement a ...

C. Preliminary Data

At this conceptual stage, extensive data collection is ongoing. However, preliminary analyses have yielded promising insights. For instance, we have observed that ... (insert any early findings or pilot data that support the feasibility of the proposed research). These initial results provide a strong foundation for the proposed specific aims and suggest that our approach is viable.

D. Research (Experimental) Design and Methods

D.1

To

D.2 The TAT-WGT Architecture

: As illustrated in Figure 1,

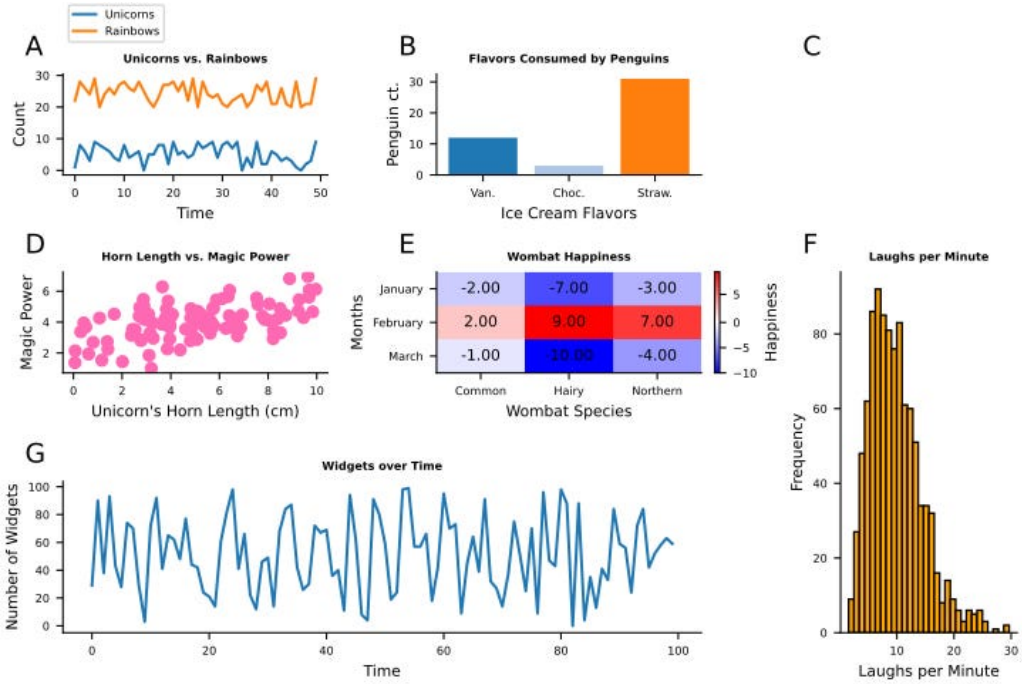

Figure 1: This is a simple figure. Replace this caption with a

D.3 Potential Difficulties and Alternative Approaches

A potential limitation is the computational bottleneck of processing large batches of 1M tokens even with hierarchical design. As an alternative approach, we will evaluate linear attention variants or State Space Models (e.g., Mamba) to replace standard Self-Attention within the WGT if $\mathcal{O}(N^2)$ complexity proves prohibitive.

D.4 Timetable

- **Month 1-2:** Data preprocessing and Hard-coded embedding construction.
- **Month 3-4:** TAT-AdaLN implementation and intra-TAD validation.
- **Month 5-6:** WGT-GRL integration, cross-ancestry evaluation, and interpretability analysis.

E. Bibliography or Reference List

[Bibliography will be inserted here via Zotero integration or edit by hand.]